Q1

1.	(a) The glass cup absorbs thermal energy (from the hot water) / Heat is lost to the glass cup.	1A	
		1	
	(b) $4200 \times 0.4 \times (100 - 93) = H \times (93 - 23)$	1M	
	$H = 168$ (J °C^{-1}) ≈ 170 (J °C^{-1})	1A	
		2	
	(c) (i) Infra-red (radiation) / thermal radiation	1A	
		1	
	(ii) Remote control / infra-red (IR) thermometer / night vision camera	1A	
		1	

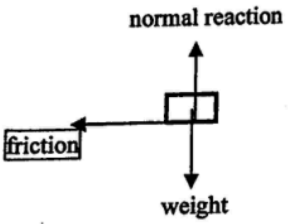
Q2

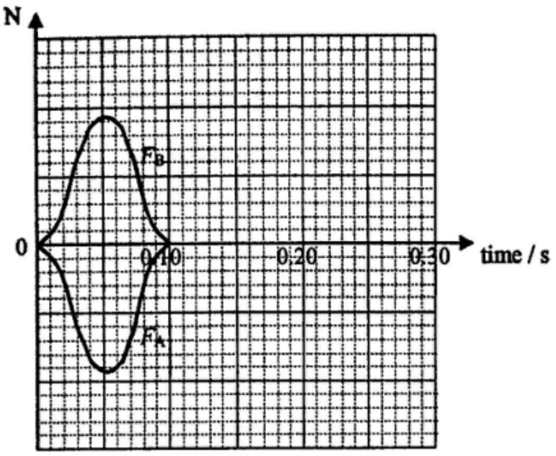
2.	(a) $s = ut + \frac{1}{2}at^2$		Or $v = u + at$, $0 = 5 + (-9.81)(t/2)$ $t = 1.02$ s
	$0 = 5t + \frac{1}{2}(-9.81)t^2$	1M	
	$t = 1.0194$ s ≈ 1.02 s or 0 s (rejected)	1A	Accept: $t = 1.00 \sim 1.02$ s
		2	
	(b) speed of the bead at its maximum height = 2 m s^{-1}		
	K.E. $= \frac{1}{2} \times 0.01 \times 2^2$	1M	
	$= 0.02$ J	1A	
		2	
	(c) The bead and the trolley have the same horizontal velocity	1A	
	and thus travel the same horizontal distance (within the same time interval t).	1A	
		2	

Q3

3. (a) The object is moving to the left with constant deceleration and (momentarily) stops at $t = 4$ s.	1A 2	
(b) Displacement = total area of the graph $= \frac{1}{2} \times (-10) \times 4 + \frac{1}{2} \times 15 \times (10-4)$ $= 25$ m Position of the object is 25 m from point O on the right side.	1M 1A 2	Or $v^2 = u^2 + 2as$ $15^2 = (-10)^2 + 2 \times \frac{15-(-10)}{10-0} \times s$ $s = 25$ m
(c) Net force acting on the object $= ma$ $= 0.4 \times \frac{15-(-10)}{10-0}$ $= 1$ N (towards right)	1M 1A 2	

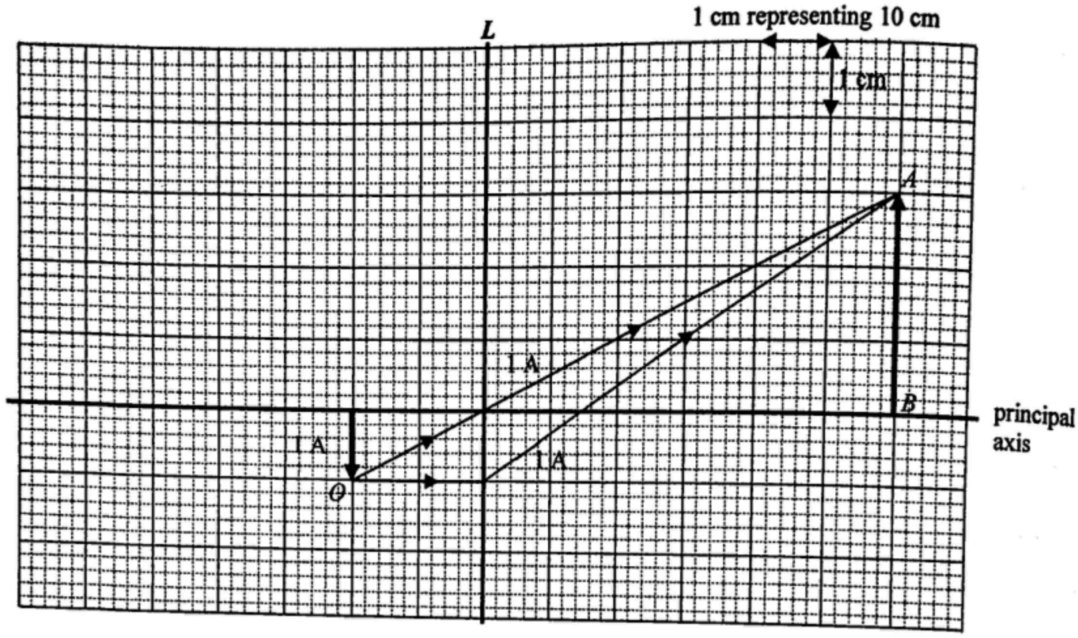
Q4

4. (a) $\omega = \frac{2\pi}{60}$ $= \frac{\pi}{30}$ or $0.105\ (\text{rad s}^{-1})$	1A 1	
(b) 	1A 1	Accept: adhesive force
(c) The required centripetal force $F_C = m\omega^2 r$, where r is the distance of plasticine Q from the clock's centre. As m and ω are constant, smaller r gives smaller centripetal force F_C. The claim is correct.	1A+1A 2	

towards the right taken as positive), give trolley *A* a push and let it collide head-on with trolley *B*.	1A	
Force-time graphs: 	2A	
The force acting on (the less massive) trolley *B*, i.e. F_B, is always equal (in magnitude and opposite in direction) to that acting on trolley *A*, i.e. F_A, as they are an action and reaction pair / according to Newton's 3rd law.	1A	
	4	

		Answer	Marks	Remarks
6.	(a)	Convex lens because only a convex lens can form a real image.	1A 1A	Or the image can be captured on a screen / the image is formed on the opposite side of the lens / only convex lens can form a magnified image.
			2	

(b) (i)(ii)

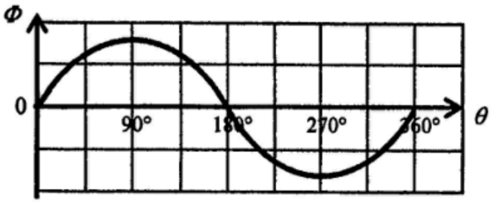
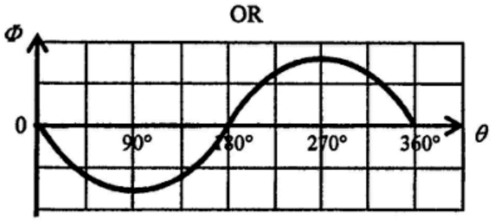


		Answer	Marks	Remarks
			3	
	(iii)	Ray diagram method: $f = 15$ cm	1A	Accept: 14 cm ~ 16 cm Or By $\frac{v}{u} = 3$, $\frac{0.6}{u} = 3$, i.e. $u = 0.2$ m By $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$, $\frac{1}{f} = \frac{1}{0.2} + \frac{1}{0.6}$ $f = 0.15$ m or 15 cm
			1	
	(c)	Move / adjust the support / smartphone closer to lens L.	1A	
			1	
	(d)	The box should be painted black inside.	1A	
			1	

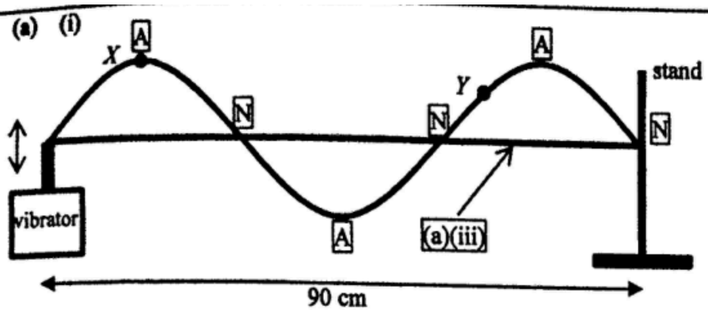
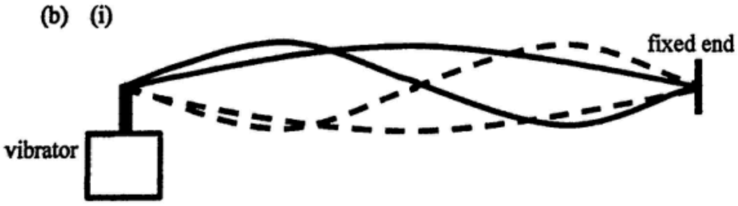
Q7

7.	(a) To protect the ammeter / limiting the current in the circuit / ensuring the circuit does not over-heat.	1A	
		1	
	(b) $R = \frac{V}{I} = \frac{3.5}{0.7}$	1M	
	$= 5.0\,\Omega\,(> 4.7\,\Omega)$	1A	
	The extra resistance (0.3 Ω) may be due to the resistance of the ammeter / contact resistances / internal resistance of the power supply.	1A	
		3	
	(c) $I = 0.25$ A at $l = 30$ cm		Accept: $I = 0.24$ A ~ 0.26 A
	$\Delta R = R_{AC} = \frac{3.5}{0.25} - 5.0$	1M	e.c.f. from (b)
	$= 9.0\,\Omega$	1A	Accept: $\Delta R = 8.5\,\Omega \sim 9.6\,\Omega$
		2	

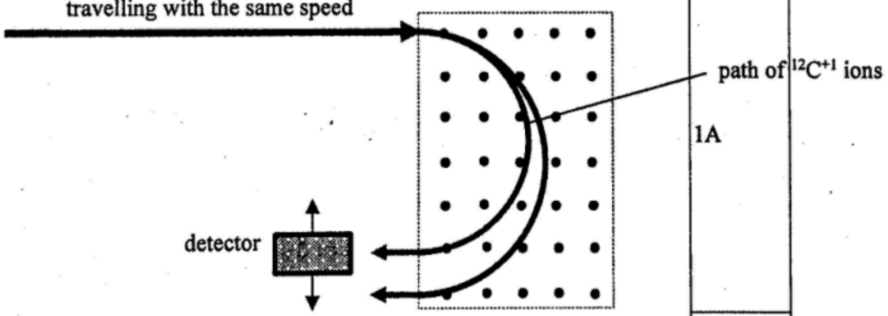
Q8

8.	(a)  OR 	2A	
		2	
	(b) (i) $\phi = BA\cos 0° = 2.70\times10^{-4}$		
	$B\times(9.00\times10^{-4})\times1 = 2.70\times10^{-4}$	1M	
	$B = 0.300\,(\mathrm{T})$	1A	
		2	
	(ii) $\varepsilon = \left\| -N\frac{\Delta\phi}{\Delta t} \right\|$		
	$= \left\| -100\times\frac{2.70\times10^{-4}}{0.01} \right\|$	1M	
	$= 2.70\,\mathrm{V}$	1A	Accept: −2.70 V
		2	

9. (a) (i) $pV = nRT$		
$1.01\times10^{5}\times V = 1\times8.31\times273$	1M	
$V = 2.2462\times10^{-2}\ \text{m}^3 \approx 2.25\times10^{-2}\ \text{m}^3$	1A	Accept: 0.0220 ~ 0.0225 m^3
	2	
(ii) Number of molecules in 1 m^3		
$= \dfrac{N_A}{\text{volume of one mole of molecule}}$		
$= \dfrac{6.02\times10^{23}}{2.246\times10^{-2}}$	1M	e.c.f. from (a)(i)
$= 2.680120\times10^{25} \approx 2.68\times10^{25}$	1A	Or $pV = nRT$ $1.01\times10^{5}\times1 = n\times8.31\times273$ $n = 44.5$ $N = 44.5\times6.02\times10^{23}$ $= 2.68\times10^{25}$
	2	
(b) $pV = \left(\dfrac{N}{N_A}\right)RT$		
$2.75\times10^{-10}\times1 = \left(\dfrac{1.00\times10^{11}}{6.02\times10^{23}}\right)\times8.31\times T$	1M	
$T = 199.2178\ (\text{K}) \approx 199\ (\text{K})$	1A	Accept: 199 ~ 200 K
	2	

10. (a) (i) 	2A	
	2	
(ii) Particle X vibrates with a larger amplitude than particle Y while they are vibrating in phase.	1A 1A	
	2	
(iii) Correct sketch of a straight string	1A	
	1	
(iv) Original wavelength $\lambda_1 = \frac{0.90}{1.5} = 0.60\,\text{m at } f_1 = 100\,\text{Hz}$ For the next stationary wave pattern, $\lambda_2 = \frac{0.90}{2} = 0.45\,\text{m}$ By $v = f\lambda$, $100 \times 0.60 = f_2 \times 0.45$ $f_2 = 133.33\,\text{Hz} \approx 130\,\text{Hz}$	1M 1A	Or $f_2 = 100 \times \frac{4}{3}$ $= 133.33\,\text{Hz} \approx 130\,\text{Hz}$
	2	
(b) (i) 	1A	
	1	
(ii) The relative strengths and proportions of the harmonics determine the unique sound characteristics of different instruments, when these harmonics superimpose and combine to produce different waveforms.	1A 1A	
	2	
(iii) Any TWO: stationary vs travelling transverse vs longitudinal energy confined in a string vs energy being transferred through air	2A	
	2	

$= 1.44 \times 10^{14}$ J $\approx 1.4 \times 10^{8}$ (MJ)	2	
(ii) $E_{out} = (680 \times 10^6) \times (24 \times 3600)$	1M	
$= 5.875 \times 10^{13}$ J $\approx 5.9 \times 10^7$ (MJ)	1A	
	2	
(iii) Efficiency $= \dfrac{E_{out}}{E_{in}} \times 100\%$		
$= \dfrac{5.88\times10^{13}}{1.44\times10^{14}} \times 100\%$	1M	e.c.f. from (a)(i)(ii)
$= 40.8\% \approx 41\%$	1A	Accept: 41% ~ 42%
	2	
(b) (i) $P = VI$	1M	
680 MW = 400 kV × I	1A	
I = 1700 A = 1.7 kA	2	
(ii) Power loss $= I^2R$	1M	e.c.f. from (b)(i)
$= (1.7 \times 10^3)^2 \times (2.90 \times 10^{-8} \times 160 \times 10^3)$	1A	
$= 13410$ W ≈ 13.4 (kW)	2	
(iii) Only a.c. voltage can be stepped up (or stepped down) using a transformer, which is cost effective.	1A	
High voltage transmission has low transmission current and so reduces power loss significantly.	1A	
	2	

12. (a) (i) $^{14}_{6}C \rightarrow {}^{14}_{7}N + {}^{0}_{-1}e$ (or β)	1A	
	1	
(ii) $k = \dfrac{\ln 2}{t_{\frac{1}{2}}}$		
$= \dfrac{\ln 2}{5370}$	1M	
$= 1.2097 \times 10^{-4}$ (yr^{-1}) $\approx 1.21 \times 10^{-4}$ (yr^{-1})	1A	
	2	
(iii) $C = C_0 e^{-kt}$		
$6.0 = 13.6\, e^{-(1.21\times 10^{-4})t}$	1M	e.c.f. from (a)(ii)
$t = 6764.7 \approx 6800$ (years)	1A	Accept: 6760 ~ 6820 (years)
		Or $(13.6)\left(\frac{1}{2}\right)^n = 6.0$ $n \log\left(\frac{1}{2}\right) = \log\left(\frac{6.0}{13.6}\right)$ $n = 1.181$ $t = n \times 5730 = 6764.7$ (years) ≈ 6800 (years)
	2	
(b) (i) An isotope is a variation of an element that possesses the same atomic number but a different mass number.	1A	Or same number of protons but different number of neutrons
	1	
(ii) $^{12}C^{+1}$ and $^{14}C^{+1}$ ions travelling with the same speed 	1A	
	1	
(iii) As the magnetic force induced on a carbon ion is always perpendicular to its velocity / direction of motion,	1A	
no work is done on the ions.	1A	
	2	